

Joseph Lin

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EDUCATION

Rutgers University

Sep 2021 – Aug 2025

Bachelor of Science in Computer Science

New Brunswick, NJ

Relevant Coursework: Data Structures & Algorithms, Software Methodology, Operating Systems, Computer Networks, Machine Learning, Database Management, System Design

EXPERIENCE

Horizon Blue Cross Blue Shield of New Jersey

Jun 2024 – Dec 2024

Software Engineer Intern, Data Infrastructure

Newark, NJ

- Designed **CI/CD pipelines** across Microsoft SQL Server, Oracle, and MongoDB for **3+** enterprise applications, replacing manual releases with standardized, zero-touch SSDT workflows built alongside the release engineering team
- Cut nightly batch processing time by **30%** on production claims data through SQL query and index optimization
- Automated routine database maintenance with **Python** and built a **Tableau dashboard** tracking **15+** database health metrics, flagging issues before they impacted production

Rutgers Blueprint Fellowship

Sep 2023 – Dec 2023

Software Engineer Fellow

New Brunswick, NJ

- Architected a Flask REST API for task management and event coordination, serving **150+ students** across **5 campus organizations**, implementing role-scoped CRUD endpoints for secure access control
- Passed a fellowship security review by implementing **JWT authentication**, token refresh flows, and **3-tier** role-based access control (admin, student, guest), validated by a PyTest suite covering auth and CRUD endpoints
- Accelerated API response times by **~40%** under concurrent load through optimized SQLite queries, implementing connection pooling, and adding prepared-statement caching

PROJECTS

Cruxly – AI Climbing Mobile App | TypeScript, Python, React Native, Supabase Auth, PostgreSQL, YOLOv8-seg, Roboflow

- Built a mobile app that scans indoor climbing walls and detects holds in real time, reaching **~0.80 mAP** on instance segmentation by fine-tuning **YOLOv8-seg** and **RF-DETR** models on a custom dataset of **1,700+** images
- Streamed segmented hold-detection results to the mobile client in **under 5 seconds** by engineering a **Node.js API** handling image uploads via **Supabase Storage** and triggering async YOLOv8 inference

Rallo – Event Coordination Mobile App (rallo.cc) | TypeScript, React Native, Expo, Supabase Realtime, PostgreSQL, Twilio

- Developed a zero-account event app that lets groups RSVP and coordinate, propagating RSVP state to all connected clients in **under 1 second** via **Supabase Realtime** subscriptions, eliminating server-side polling
- Restricted event visibility and RSVP access to invited participants only by structuring **PostgreSQL** schemas with **Row-Level Security**

Restaurant Full-Stack Ordering Platform | JavaScript, React, Next.js, Tailwind CSS, Hygraph CMS

- Processed **150+ orders** over **3 months** in live production for my family's Chinese restaurant, replacing Grubhub and Uber Eats, and saving **\$500+/month** in commission fees
- Gave staff real-time order visibility from submission through pickup by rendering **100+ menu** items via **Next.js** server-side rendering and **GraphQL** from **Hygraph CMS**, modeling the order lifecycle as a state machine

NBA Statistics REST API | Java, Spring Boot, PostgreSQL, Spring Data JPA

- Constructed a **Java REST API** serving multi-season stats from **150+ NBA players** using **Spring Boot** and **PostgreSQL**, designed for sports analytics and fantasy sports applications
- Modeled relational data across players, teams, and games using **JPA/Hibernate** with PostgreSQL, implementing server-side pagination, filtering, and multi-season sorting via **Spring Data JPA**

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, HTML/CSS

Frameworks and Libraries: React/React Native, Next.js, Node.js, GraphQL, Flask, Spring Boot, Pandas, NumPy, OpenCV

Databases: PostgreSQL, Microsoft SQL Server, MySQL, SQLite, MongoDB, Oracle

Tools and Platforms: Docker, Git, Supabase, Vercel, Expo, Tableau, Roboflow, Figma, Jira, Jupyter

Concepts: RESTful API Design, CI/CD, Agile/Scrum, JWT Auth, Full Stack/Mobile Development, Computer Vision, System Design